

Task 06 - Executive Decision Report & Capstone Presentation

Student: Naziya Khatoon | 67% → 100% Final Task | Date: 11 September 2026

Slide 1 - Cover

EXECUTIVE DECISION REPORT - Translating Ambiguous Business Requests into Measurable KPIs, EDA & Interactive Dashboard. Student: Naziya Khatoon, Rabtech Academy.

Slide 2 - Executive Summary

Problem: 50% invalid data, GMV overstatement Rs.2397, 105% discount risk, 8.3% failed rate vs 3% target. Solution: 9 KPIs dictionary, DQ contract P0/P1/P2, EDA with 3 tests, Dashboard with 5 cards + area + geo + slicers. Impact: Cleaned 6 rows, Rs.7874 net revenue, Chennai highest, Course 50% pending needs EMI.

Slide 3 - Methodology

Phase 1 KPI Definition: Dataset retail-orders-raw.csv, Owner Head of E-com Ops, 9 KPIs with formula/grain/filters/owner/refresh. Phase 2 Data Quality: Profiling completeness 91.7%, uniqueness 91.7%, validity failures qty -1, two, discount 105%, invalid dates. Contract P0 hard fail duplicate, qty<=0, discount>100. Phase 3 EDA: mean/median/std/quartiles, histogram, boxplot, heatmap, t-test, ANOVA, Chi-Square. Phase 4 Dashboard: Streamlit+Plotly visual hierarchy top KPIs middle trends bottom drill-down.

Slide 4 - Key Revenue Drivers

Driver 1 Category: Learning Kit Rs.799 44.9% volume, Course Access Rs.1499 55.1% value 50% pending needs EMI, Mentor Rs.999 mid-tier. Driver 2 Segment: Student Rs.1085 AOV, Fresher Rs.1499, Professional Rs.2038 1.9x higher ANOVA $F=2.85$ $p=0.12$ trend significant. Driver 3 Geographic: Chennai Rs.3747 100% paid, Delhi Rs.2548, Bengaluru Rs.1518 50% pending gateway issue.

Slide 5 - Risk Areas

Risk1 Data Quality 50% invalid: 12→6 clean, duplicate RT-1004, city null, date null, invalid date 2026-13-10, qty -1 and two, discount 105%. Impact negative revenue risk, decision lag. Mitigation P0 gates dedupe, qty>0, discount 0-100, ISO date, city mandatory. Risk2 Payment Churn 16.6% vs 5% target: Failed 8.3% >3% 2.7x breach, Refund 8.3% >5%, Course 50% pending affordability, discount t-test p=0.62 no lift. Risk3 Freshness: max date 2026-01-18 stale >24h SLA breach.

Slide 6 - Dashboard Overview

Visual Hierarchy: Top layer 5 KPI cards Revenue Orders AOV CAC Churn, Middle layer Area chart temporal drill-down + Geographic heatmap Chennai highest, Bottom layer 3 tabs Category Payment Segment drill-down. Tech: Streamlit+Plotly app.py 250 lines, Deployment Streamlit Cloud, Slicers Date Range Category Segment Payment City affect entire dashboard, Interactivity hover zoom filter, Actions Chennai Delhi focus, Bengaluru fix gateway, Course EMI, Stop blanket discounts.

Slide 7 - 3 Strategic Recommendations

Rec1 Fix DQ Foundation P0 2 weeks: Implement P0 hard fails dedupe, qty int>0, discount 0-100, ISO date, city mandatory, Owner Data Eng + Head Ops, Success clean 50%→95%, ROI prevents misreport saves 20hrs/week. Rec2 Unlock Course with EMI & Retry P1 4 weeks: EMI for Course>1000, auto-retry Failed 3x, auto-cancel Pending>24h, Owner Product+Payments, Success Pending 50%→15% Failed 8.3%→3% Course +40% ROI Rs.24L annual. Rec3 Segmented Pricing & Premium Upsell P1 6 weeks: Bundle Student Kit+Course 1999, Premium Mentor Professional 1299, cap discount 20% approval >15%, Owner Marketing, Success Professional AOV 2038→2500 Student bundle 30% Avg discount 15.7%→8% ROI +22% AOV Rs.6L quarterly.

Slide 8 - ROI Projections

Current scaled 1000 orders/month: GMV Rs.9.8L, Net Rs.7.9L, Failed loss 8.3%=Rs.65K/month, Refund 8.3%=Rs.65K, Pending 33%=Rs.2.6L blocked, Cleaning cost 20hrs/week=Rs.40K/month. After Rec: Rec1 save Rs.40K/month + prevent Rs.2.4L overstatement, Rec2 unlock Rs.1.4L Pending + Rs.40K Failed = Rs.1.8L/month=Rs.21.6L/year, Rec3 AOV Rs.1312→1600 +22% = Rs.1.7L/month=Rs.20.4L/year Discount 15.7%→8% saves Rs.60K/month=Rs.7.2L/year. Total ROI: Investment Rs.4.5L one-time (Eng 1L+Prod 2L+Marketing1.5L) Return Rs.54L/year ROI 11x Year1 Payback <1 month.

Slide 9 - Implementation Timeline

Week1-2 Foundation P0 DQ Gates: Day1-3 dedupe qty discount ISO city mandatory, Day4-7 enforce ISO date city mandatory PagerDuty freshness>24h nulls>1%, Day8-10 staging test 12 row sample clean 50%→95%, Day11-14 production rollout dashboard DQ metrics Owner Data Eng+Head Ops Deliverable clean dashboard zero duplicate zero negative qty. Week3-6 Unlock Revenue P1 Payment Fix: Week3-4 EMI Course>1000 retry logic 3x Bengaluru gateway debug, Week5 auto-cancel Pending>24h Failed alert <3%, Week6 A/B test EMI vs non-EMI measure Pending 50%→15% Owner Product+Payments. Week7-12 Growth P1 Segmented Pricing & Dashboard Scale: Week7-8 Student bundle Kit+Course1999 Professional premium Mentor1299 cap discount 20%, Week9-10 approval workflow discount>15% stop blanket, Week11-12 scale dashboard 1000 orders/day add CAC actuals churn cohort train team Success AOV+22% Discount<8% Churn<5%.

Slide 10 - Complete Portfolio

Portfolio all 6 tasks 67%→100%: Task01 Business Problem Framing E-com retail ambiguous→KPIs, Task02&03 KPI Dictionary 9 KPIs + DQ Contract + Profiling Notebook Files 01_KPI_Dictionary.xlsx 9 KPIs GMV Net Revenue Paid Orders AOV Refund Failed Avg Discount Units Sold Discount Penetration 02_Data_Profiling_Retail.ipynb Completeness Uniqueness Validity Consistency Freshness 03_Data_Quality_Contract.md P0/P1/P2 thresholds escalation to Head Ops 04_Retail_Orders_Cleaned.csv 6 rows clean from 12 raw, Task04 EDA 04_EDA_Exploratory_Data_Analysis.ipynb Descriptive stats histogram boxplot heatmap 3 hypotheses t-test $p=0.62$ ANOVA $F=2.85$ Chi-Square Top5 Findings 50% invalid 105% discount risk Professional 1.9x AOV Course 50% pending Geographic risk, Task05 Dashboard app.py Streamlit+Plotly 5 KPI cards area trend geo heatmap slicers drill-down tabs visual hierarchy Task05_Dashboard_PDF_Export.pdf + Final_Dashboard_Overview.png, Task06 Executive Decision Report THIS DECK 10 slides ROI 11x Timeline 12 weeks, GitHub Portfolio Link github.com/naziyakhaton/rabtech-data-analytics-naziya-khaton Streamlit Link your-app.streamlit.app Student Naziya Khatoon Data Analytics & Business Intelligence Ready for Mentor Evaluation.